

MEHANK

AI & Backend Engineer | B.Tech CSE | India, Mumbai

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Education

Vellore Institute of Technology

2022

B.Tech. in Computer Science, Minor in Cybersecurity and Digital Forensic

CGPA: 8.1

Work Experience

AI Engineering Intern

Dec 2025 – Present

PrudentTech IT Solution

- Architected an AI-powered voice calling system integrating Whisper (Speech-to-Text), Gemini LLM for conversational intelligence, and Azure Text-to-Speech for real-time voice synthesis.
- Implemented end-to-end conversational pipeline including audio processing, prompt orchestration, LLM response handling, and automated testing, improving call response accuracy and reducing manual intervention.
- Collaborated with engineering team to optimize latency and ensure stable deployment through structured testing and debugging workflows.

AI Engineering Intern

Oct 2025 – Nov 2025

Shell plc (Public Limited Company)

- Built and operationalized Python-based machine learning data pipelines processing large-scale datasets, achieving 99.9% system uptime through automated testing (PyTest) and CI/CD integration.
- Collaborated with cross-functional teams to standardize deployment workflows and authored technical documentation, improving release reliability and reducing onboarding time by ~30%.

Frontend Developer Intern

May 2025 – July 2025

Sahoo Engineering & Contractors

- Developed a secure production-grade React application using **TypeScript** strict mode and **Zod validation** to enforce input integrity.
- Improved frontend performance to **60 FPS** by optimizing rendering logic and removing unused assets.

Projects

RAG Document Q&A System— Flask, ChromaDB, Ollama, NLP, LangChain

[View project](#)

Jan 2026

- Designed a Retrieval-Augmented Generation (RAG) system to process 1,000+ documents using embedding-based semantic search, reducing manual lookup time by 40% and improving contextual response accuracy by ~30%.

Legacy Booking System Security— Laravel, PHP, MySQL, Next.js, RESTful API

[View project](#)

Dec 2025

- Migrated a legacy PHP system to Laravel, resolving 15+ SQL injection vulnerabilities and securing REST APIs, reducing application security risks by ~60%.

Rooftop Images Segmentation — Python, PyTorch, CI/CD, TensorFlow

[View project](#)

Nov 2025

- Deployed a deep learning segmentation pipeline to process 1,600+ images, automating evaluation workflows and reducing manual review effort by 80%.

SuPOR Cipher Suite — Python, NumPy, Modular Architecture

[View project](#)

May 2025

- Created a stream cipher in Python for secure file encryption and decryption, supporting files up to 50MB.

Real-time Sign Language Recognition — Kotlin, TensorFlow Lite, YOLOv8, CameraX

[View project](#)

Feb 2025

- Engineered a real-time Android application using Kotlin and TensorFlow Lite to recognize 26+ ASL gestures, achieving 35 FPS and 80 ms latency through YOLOv8-based edge inference and optimized CameraX processing.

Packet Sniffer Network Analysis Tool — Python, Flask, Scapy

[View project](#)

Mar 2024

- Constructed a multi-threaded packet analyzer for HTTP/TLS traffic inspection, improving real-time network traffic analysis speed by ~35%.

Technical Skills

Programming & Databases: Python, Go, Java, PHP, SQL, JavaScript, PostgreSQL

Frameworks & Libraries: PyTorch, Tensorflow, FastAPI, Node.js, LangChain, PyTest, Flask, Laravel

Concepts & Tools: REST APIs, Docker, Git, CI/CD, AWS, Machine Learning, Deep Learning, LLM Applications.

Achievements

- Rank<1000 in TCS CodeVita**(Round 2 Qualifier) among 100,000+ participants.
- Earned Certifications in **Cybersecurity & DevOps (IBM)**, **Google Cybersecurity**, and **Software Engineering (HackerRank)**.
- Achieved **5-Star Gold Badge in SQL** (HackerRank) and **Linux Badge** (TryHackMe).